

Tutorial - Drawing Network Diagrams

You have been shown that network designs require multiple diagrams styles. This tutorial is intended to encourage you to test at least two diagram packages that will help you with your assignment.

There are physical diagrams that show the location of cabling as well as terminal and intermediary devices and there are logical diagrams that indicate how the network is intended to work and interconnect

There are other diagrams that become necessary in specific circumstances

Each of these diagrams may be mainly pictorial, showing higher level information, or detailed and giving very specific information. The information may relate to configuration or to installation instruction.

For example, when designing a physical classroom layout, the diagram could show a number of data points as a general statement. This would not be suitable for a final layout as there is not enough detail to be able to position the desks or datapoints. It would however allow a designer to determine the number of cables that will be required which would in turn determine the size of the containment (trunking) that would need to be installed. The “Quantity Surveyor” would be able to estimate the cabling and equipment and so a formal quotation with costs can be provided to the customer. The diagram could not confirm that the number of desks would fit the space or that there would be enough room to move around, this would need a more detailed drawing.

There are many different diagramming tools available. For networks, two very popular diagramming tools are Cisco Packet Tracer and MS Visio, though there are many others

For this tutorial you need to reproduce the two diagrams below in both Packet Tracer and Visio. You do not need to show the measurements on your work, they are provided to help you with the layout. Exact dimensions are not required for this exercise.

Diagram 1

To produce Figure 1, you will need to work in the physical tab when using Packet tracer.

You will need to open the “Walls, doors and windows” stencil from “Maps and floor plans”, “Building Plan” when using Visio.

Save your work when you have created the diagram. After saving the work, add trunking to the diagram that will connect point A to point B and point A to point C (point B to point C via point A). Your trunking should be at a realistic scale and must not cross any doorways.

Diagram 2

To produce Figure 2 you will need “network and peripherals”, “Network Locations” and “Computers and Monitors” in Visio. You do not need to add the labels shown, they are only shown for clarification.

The icons are different in Packet Tracer, you should use a 4321 router, 2950 switch and a 5505 firewall (also known as an ASA).

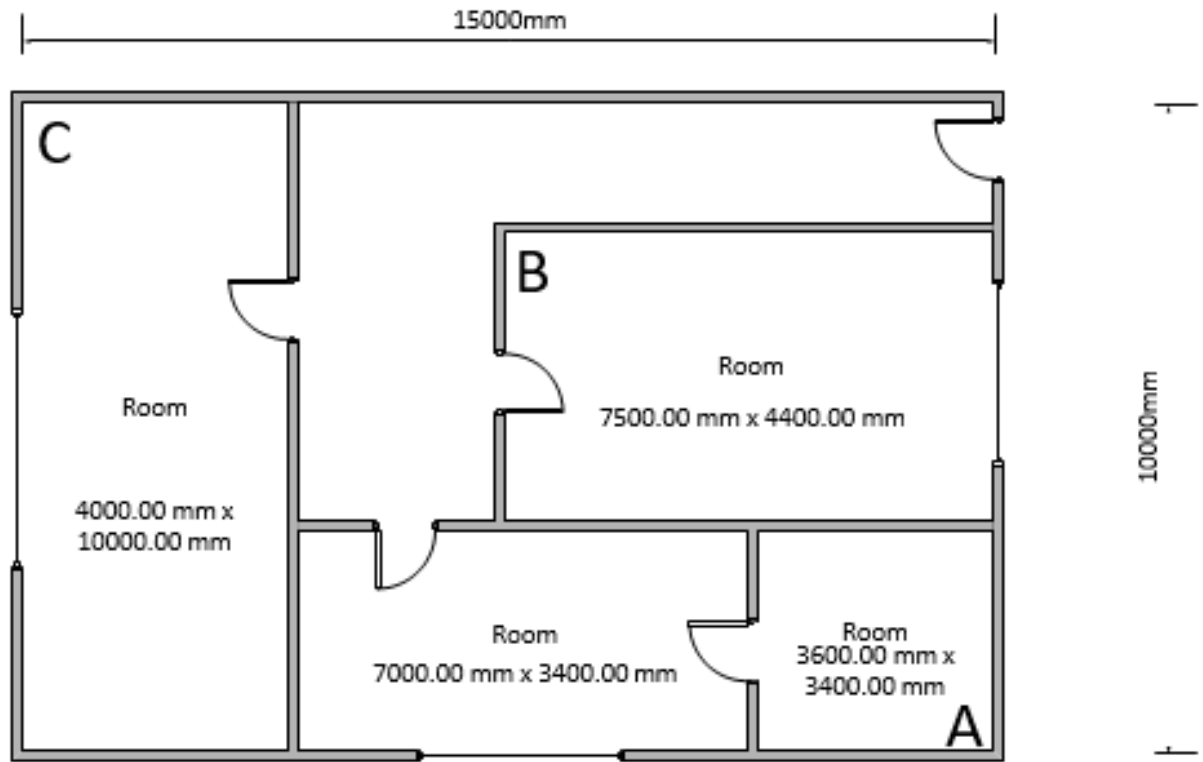


Figure 1: Physical Diagram

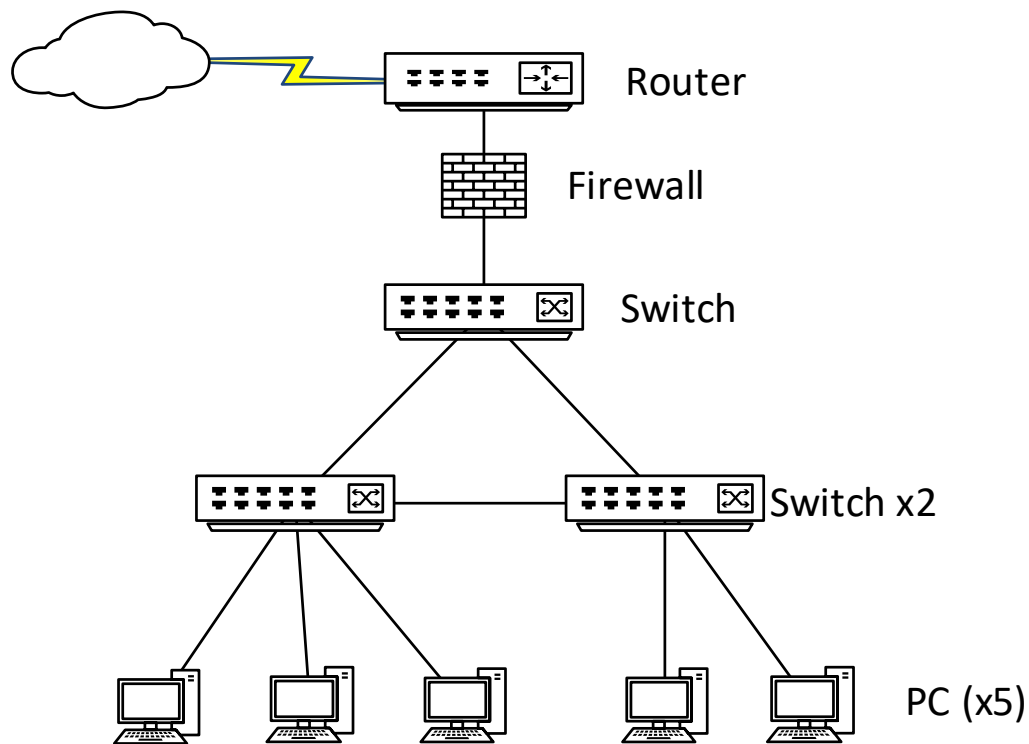


Figure 2: Logical Diagram

General comments

You should use the “connector” in visio to join devices. Make sure that you attach to a point on the device icons, this will let you move the icons on the page without breaking the link.

Some of the connection points are in the centre of the icon in Visio, this can make the diagram look untidy. The presentation can be improved by selecting the connection and using “send to back” or selecting the device and using “bring to front”

You are very likely to find that the two packages are not equal in capability or ease of use when creating diagrams. You should also be clear that they have different capabilities in terms of simulation and documentation. Packet tracer will simulate network operations while Visio can provide a basic management platform through the use of hyperlinks.